



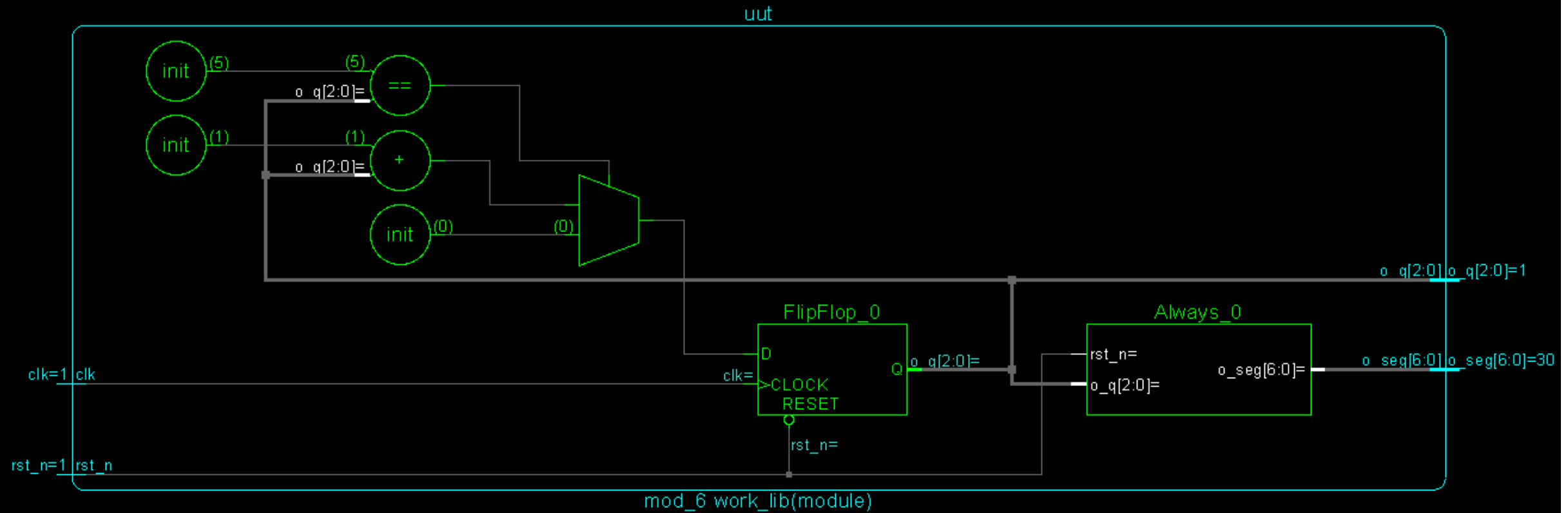
SoC 102: Design

중간점검: Basic Sequential Logic Circuit / Synthesis / PNR

MOD-6

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Design: Schematic



Design: RTL Coding

```
`timescale 1ns / 1ps

module mod_6 (
    // port list
    clk
    rst_n
    o_q
    o_seg
);

    // port declaration
    input      clk      ;
    input      rst_n    ;
    output [2:0] o_q     ;
    output [6:0] o_seg  ;

    // modeling
    reg [2:0] o_q      ;
    reg [6:0] o_seg    ;
    always @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            o_q <= 3'b000;
        end else begin

            if (o_q == 3'b101) begin
                o_q <= 3'b000;
            end else begin
                o_q <= o_q + 1'b1;
            end
        end
    end

    // 7_seg Decoder Logic
    always @(*) begin
        if (!rst_n) begin
            o_seg = 7'b0000000;
        end else begin
            case (o_q)
                3'b000 : o_seg = 7'b1111110; //0
                3'b001 : o_seg = 7'b0110000; //1
                3'b010 : o_seg = 7'b1101101; //2
                3'b011 : o_seg = 7'b1111001; //3
                3'b100 : o_seg = 7'b0110011; //4
                3'b101 : o_seg = 7'b1011011; //5
                default : o_seg = 7'bz;      //disabled
            endcase
        end
    end

endmodule
```

Testbench

```
`timescale 1ns / 1ps

module tb_mod_6();

// port list
reg          clk          ;
reg          rst_n        ;
wire [2:0]    o_q          ;
wire [6:0]    o_seg        ;

// DUT instantiation
mod_6 uut (
    .clk          (clk          ),
    .rst_n        (rst_n        ),
    .o_q          (o_q          ),
    .o_seg        (o_seg        )
);

// clock gen
always #5 clk = ~clk;

initial begin

    #10;
    clk      = 0;
    rst_n    = 0;

    #10;
    rst_n    = 1;

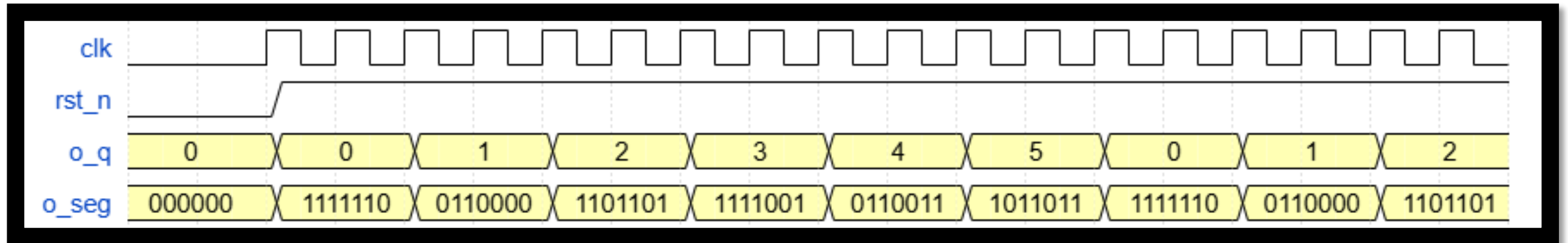
    #200;

    $finish;

end

endmodule
```

Design: Waveform



Synthesis

```
module mod_6(clk, rst_n, o_q, o_seg);
    input clk, rst_n;
    output [2:0] o_q;
    output [6:0] o_seg;
    wire clk, rst_n;
    wire [2:0] o_q;
    wire [6:0] o_seg;
    wire n_0, n_1, n_2, n_3, n_4, n_5, n_6, n_7;
    wire n_8, n_9, n_10, n_11, n_12, n_13, n_14, n_15;
    wire n_18, n_20, n_23;
    TBUFXL g7_2398(.A (n_20), .OE (n_23), .Y (o_seg[6]));
    TBUFXL g2_5107(.A (n_11), .OE (n_23), .Y (o_seg[5]));
    TBUFXL g8_6260(.A (n_15), .OE (n_23), .Y (o_seg[4]));
    TBUFXL g9_4319(.A (n_20), .OE (n_23), .Y (o_seg[3]));
    TBUFXL g12_8428(.A (n_14), .OE (n_23), .Y (o_seg[0]));
    NOR2BX1 g331_5526(.AN (n_10), .B (n_7), .Y (n_18));
    TBUFXL g11_6783(.A (n_9), .OE (n_23), .Y (o_seg[1]));
    TBUFXL g10_3680(.A (n_3), .OE (n_23), .Y (o_seg[2]));
    OAI21XL g334_1617(.A0 (n_6), .A1 (n_13), .B0 (n_12), .Y (n_15));
    OAI22XL g335_2802(.A0 (n_2), .A1 (n_13), .B0 (n_8), .B1 (n_12), .Y
        (n_14));
    OAI21XL g333_1705(.A0 (o_q[0]), .A1 (n_12), .B0 (n_13), .Y (n_11));
    OAI22XL g329_5122(.A0 (n_5), .A1 (n_13), .B0 (n_10), .B1 (n_12), .Y
        (n_20));
    AOI21XL g339_8246(.A0 (o_q[0]), .A1 (n_8), .B0 (n_12), .Y (n_9));
    AOI21XL g340_7098(.A0 (o_q[1]), .A1 (n_6), .B0 (n_5), .Y (n_7));
    NOR2BX1 g332_6131(.AN (n_10), .B (n_1), .Y (n_4));
    NOR2XL g336_1881(.A (o_q[0]), .B (n_13), .Y (n_3));
    OR3X1 g337_5115(.A (n_0), .B (n_8), .C (n_2), .Y (n_23));
    NOR2X1 g343_7482(.A (o_q[1]), .B (n_6), .Y (n_5));
    NAND2XL g345_4733(.A (rst_n), .B (n_8), .Y (n_13));
    AOI21X1 g338_6161(.A0 (o_q[1]), .A1 (o_q[0]), .B0 (o_q[2]), .Y
        (n_1));
    NAND2XL g344_9315(.A (rst_n), .B (n_2), .Y (n_12));
    NAND2X1 g341_9945(.A (o_q[2]), .B (o_q[0]), .Y (n_10));
    INVXL g347(.A (rst_n), .Y (n_0));
    DFFRX1 \o_q_reg[2] (.RN (rst_n), .CK (clk), .D (n_4), .Q (o_q[2]),
        .QN (n_8));
    DFFRX1 \o_q_reg[1] (.RN (rst_n), .CK (clk), .D (n_18), .Q (o_q[1]),
        .QN (n_2));
    DFFRX1 \o_q_reg[0] (.RN (rst_n), .CK (clk), .D (n_6), .Q (o_q[0]),
        .QN (n_6));
endmodule
```

```
set sdc_version 2.0

set_units -capacitance 1000ff
set_units -time 1000ps

# Set the current design
current_design mod_6

create_clock -name "clk" -period 10.0 -waveform {0.0 5.0} [get_ports clk]
set_clock_transition 0.1 [get_clocks clk]
set_clock_gating_check -setup 0.0
set_wire_load_mode "enclosed"
set_clock_uncertainty -setup 0.01 [get_ports clk]
set_clock_uncertainty -hold 0.01 [get_ports clk]
```

Mod_6_final.sdc – Synopsys Design Constraints

Mod_6_netlist.v – Gate Level Netlist

```
***** End: VERIFY CONNECTIVITY *****
Verification Complete : 0 Viols. 0 Wrngs.
(CPU Time: 0:00:00.0 MEM: 0.000M)
```

자기평가

	수강여부	숙련도(상/중/하)
회로이론		
전자회로	o	하
디지털논리회로	o	하
디지털 시스템설계		
신호와 시스템		
HDL 설계언어(Verilog/VHDL)		
컴퓨터구조론		
디지털통신공학		
VLSI 설계		
C언어	o	하
마이크로프로세서프로그래밍		

- 본전공 → 반도체설계과
- 직무희망 / 관심영역 → ANALOG LAYOUT / PD
- 방학 학습계획 → (유튜브) 뽕교수 강의
→ Digital Design
→ Digital Systems Design Using Verilog
- Git 주소 → <https://github.com/ehdtjq1289>